

Problem 11

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Problem 1 Compute the integral: $\int \frac{x}{\sqrt{x-4}} dx$.

Explanation. There are two possible solutions: Let $w = \sqrt{x-4}$. Then

$$dw = \frac{dx}{2\sqrt{x-4}} \implies 2dw = \frac{dx}{\sqrt{x-4}}$$

and

$$w = \sqrt{x-4} \implies x = w^2 + 4.$$

Substituting these into the original integral and evaluating:

$$\begin{aligned} \int \frac{x}{\sqrt{x-4}} dx &= \int 2(w^2 + 4) dw \\ &= 2 \left(\frac{w^3}{3} + 4w \right) + C \\ &= 2 \left(\frac{(\sqrt{x-4})^3}{3} + 4\sqrt{x-4} \right) + C. \end{aligned}$$

The other solution: Let $v = x - 4$. Then $dv = dx$ and $x = v + 4$.

$$\begin{aligned} \int \frac{x}{\sqrt{x-4}} dx &= \int \frac{v+4}{\sqrt{v}} dv \\ &= \int (v^{1/2} + 4v^{-1/2}) dv \\ &= \frac{2}{3}v^{3/2} + 8v^{1/2} + C \\ &= \frac{2}{3}(x-4)^{3/2} + 8(x-4)^{1/2} + C \end{aligned}$$